

ELEE 4230/6230: Course Project

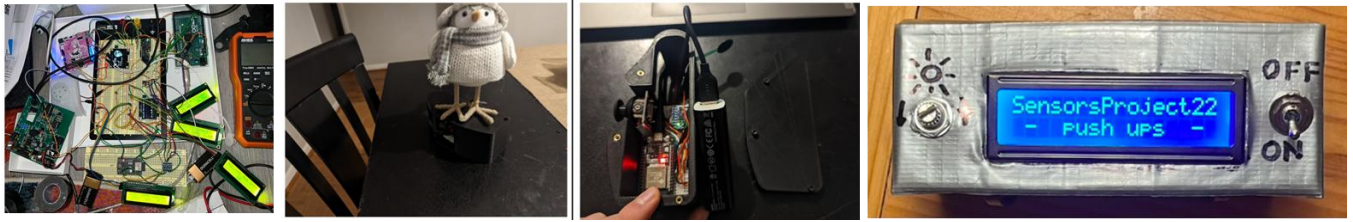


Figure 0: Past Projects. The finished project should be a Rev 1.

Why

As part of an effort to make this class more hands on and directly relevant to your life, we will have an ongoing project throughout the course. I have found that when engineers can see sensors in daily life it's often pretty motivating to figure out how they work. I also think engineers should have an opportunity to create and problem solve, and I hope you'll use this opportunity to build something meaningful in your life.

When

Per the course schedule, we have ongoing updates summarized in the table. Given these deadlines are known at the start of the semester, a 50% per day late deadline is applied. There are checkpoints throughout the semester to keep progress going, see the calendar.

Who

For 99.9% of the projects, I would like this to be an individual project. A few reasons why: 1) Going back to the 'why' section, each of you are in a unique situation, and so I want this to be something you are personally interested in. 2) I want to see you apply the concepts from the lectures. I have strategically placed the due dates throughout the course so that you can show the ideas from lecture manifesting in your situation. 3) Project scope. This isn't sophomore design where the entire class is a project. This isn't a true multi-disciplinary class; sure, there are multiple majors taking this class, but all are judged in the same way. I expect 99.9% of the projects to be individual, so if you think you are the 0.01% you can reach out to me, but overall, this project is intended to be individual.

Where

Well, anywhere provided you can record the data and get to ELC before the due dates.

What

The goal is a novel-to-you project that is meaningful. When you finish this project, you don't need to go apply for a patent, although that would be impressive, but you should seek to incorporate your interests. As further explained in the 'how' section you can incorporate projects from an internship or work within a club. You can also build on project from previous classes. Either way, you will need to go beyond and satisfy the requirements of this document.

Constraints

There are a few constraints for this project. Since each project is unique feel free to discuss these with your instructor:

- Each project **MUST** have a RAW sensor. This means that you do all of the signal processing to get the signal from an unconditioned state to a state acceptable for your logic processing.
 - The signal processing must be active.
 - The signal processing should have an appropriate analog filter.
- For a passing grade, each project **MUST** be completely powered by a signal battery source. *Note:* there can be a compelling reason to need wall power. However, the need to power a dual supply op-amp is not a compelling enough reason, use a 0-4.5v-9V style setup. Discuss with your instructor as needed.
- Algorithm Requirements:
 - Each sensor used must be coded such that it is a subfunction, so that when you call that function the value is returned of that sensor.
 - Each project must include an appropriate Kalman filter algorithm.
 - Each project must include an appropriate “go beyond” algorithm.
Examples include:
 - Writing an IOT program to log data.
 - Writing a merge program to merge data from different sensors (i.e. perfect cup of coffee example throughout the class).
 - Writing, not using a pre-packaged library, of another digital signal processing algorithm (i.e. FIR filter).
- Physics requirements:
 - Sensor systems must have redundancy in 1 measurement from 2 different branches of physics.
 - Sensor systems need at least a total of 3 measurements of various measurand(s) of the environment.
 - Note... Switches are nice! Actuators are nice! But these are not considered sensors.

Practical advice from years past...

- Choose a measurand you can actually go measure and test repeatedly.
- For your pre-packaged sensors (i.e. the non-raw ones) you should use ones that have a good datasheet and tell you information about them.
- DO WHAT YOU SAY YOU ARE GOING TO DO.
 - Do not tell me you are going to create an autonomous vehicle parking system then show up with a breadboard and an ultrasonic sensor. Saying, “Well I could...”.
- A “digital” temperature sensor (or humidity or ...) means that you have not analyzed it deep enough. Yes, the output is a digital signal, but there are many many steps of signal processing to get it to the I2C signal you are reading. If you cannot determine these steps, this is likely not a good sensor for this project and you will lose points not describing the signal processing path.
- DO NOT WRITE LIKE YOU ARE ON THE MARKETING TEAM.
 - I think this project is great already, you don’t need to convince me. Your report needs technical detail about the signal processing and the engineering.
- The goal is SYSTEMS engineering. Thinking in terms of inputs and outputs. What are all the inputs to your sensor? What are its outputs? Think in terms of each subsystem’s inputs and outputs.

Grading Boundaries

All projects will be required to take physical hardware, record real-life data, and document conclusions in a report. How much effort you put into realizing hardware will determine the maximum grade possible. Using the table below, you can assume your documentation is flawless, but with lack of hardware comes lack of grade.

Max Grade	Hardware	Power	Box
65%	Project is on a breadboard.	Project is powered by a power supply ultimately connected to 120VAC	No box exists
75%	Project is on a breadboard.	Project is powered by a single battery. (Unless sufficient justification for wall power).	No box exists
85%	Project is on an appropriately sized breadboard.	Same as above.	An appropriately sized box exists.
95%	Same as above except a PCB is designed.	Same as above.	Same as above.

100%	A physical PCB exists and is in use in the project.	Same as above.	Same as above.
------	---	----------------	----------------

DO NOT WAIT TO 3D PRINT. EXPECT FAB LAB DELAYS!

Additional Comments

It is your instructors hope that you'll be able to use the project to solve a real-life problem or implement a concept. Graduate students should use this to connect to their research. However, the boundaries for grading are the content inside this class. For example, you use the sensors project to measure temperature and you are going to use that information in a feedback control system to heat/cool. Your instructor is only grading the sensing aspect. So, the feedback control system does not help or hurt your grade on this project. If you have questions ask your instructor on this.

Updates

Since this is a course project, not a final project, there are deadlines throughout the course to help motive progress. However, it is acknowledged that most of the work will happen later in the semester after the bulk of the material has been covered.

<i>Task</i>	<i>Date Due</i>	<i>Description</i>
CP 1	With Lab 1c	Submit a project idea to the ELC folder. See ELC for the form.
CP 2	With Lab 2c	Submit photos/videos showing you have access to the hardware needed to complete the project.
CP 3	With Lab 3c	Submit a video showing the RAW sensor working.
Cp 4	See Course Schedule	Demonstrate the project and complete the report.

Grading Algorithm

<i>Task</i>	<i>Weight</i>	<i>Scales</i>
CP 1	2.5%	1 – good go for it 0.75 – on time but need to revise project 0.5 – late, but good project so go for it 0.25 – late, and need to revise project 0 – no submission
CP 2	2.5%	1 – Great! You have the hardware! 0.75 – You have most of the hardware. 0.5 – Great! You have the hardware but are late. 0.25 – late, and need to get more hardware. 0 – no submission
CP 3	5%	1 – Clear video. Shows signal processing and sensor working to a stimulus. 0.75 – Some aspect is not shown or missing.

		0.5 – Great! You have it all, but it was late. 0.25 – late, and need to improve some aspect of it. 0 – no submission
Cp 4	90%	A report and physical demonstration.

Final Demonstration

Please see the course schedule, time will be allocated at the end of the semester to present your sensors project. You must present in-person, unless there is a compelling reason you can't with your project. Below are sample projects who didn't present in lab:

- You want to go fishing with Dr. Trudgen to show off your course project that centers around weighing fish.
- Your project involved measuring the length of the clutch pedal of your car so you can track clutch life.
- Your project involved your massive fish tank aquarium setup.

You get the idea! Go measure something cool!

Grade Fees

- We live in such an amazing world with so many tools to help! Therefore, a 10% flat fee will be applied to all reports with hand drawn figures or equations.
- Your writing needs to be bookended. A 5% fee is applied to breaking this convention.
 - Intro/Conclusions: Ok to write in 1st person.
 - All other sections: 3rd person only.
- In order to grade the final report, a hardware demonstration will be needed.
- Changes may occur as needed throughout the semester.

Grading the final report

Your instructor has found that the best feedback comes from conversations. This happens for a few reasons, 1) the "bit rate" is so much higher. 2) Confusion points can be clarified much easier, and 3) the tone of voice is not misconstrued.

Grading will happen in this general algorithm:

- 1) Bin the projects based on the hardware level (see grading boundaries).
- 2) Read every word of the report.

Based on time needs at the end of semester will determine if a rubric or successive bit ADC style algorithm is used.

Useful Resources from the UGA Library Website:

1. A new website called <https://write.uga.edu/> exists with many resources.
2. *Scientific Writing: A Reader And Writer's Guide* by Jean-luc Lebrun
Description: An energetic and easily read book for effective technical writing.
Permalink: http://proxy-remote.galib.uga.edu/login?url=http://search.ebscohost.com/login.aspx?direct=true&db=nlebk&AN=203920&site=eds-live&ebv=EB&ppid=pp_1

Answers questions like:

- How to sustain attention and motivation (Chapters 2 and 4)
- How to share knowledge to unfamiliar audiences (Chapter 5)
- How to write an abstract (Chapter 11)
- How to write an introduction (Chapters 13 and 14)
- How to write a conclusion (Chapter 16)

3. "The Engineers' Guide to Technical Writing: Insights for Budding Engineers" by Abdul Khan
Description: A straightforward and brief article providing quick structural guidelines.
Permalink: <http://proxy-remote.galib.uga.edu/login?url=http://search.ebscohost.com/login.aspx?direct=true&db=eric&AN=EJ1239114&site=eds-live>

Answers questions like:

- Initial, Design, Interim, and Laboratory Report Guidelines
- What to include in a professional email
- Motivation for technical writing experience and expertise

4. *Technical Writing: What It Is and How to Do It* by Julie M. Zelenik
Description: A detailed book to motivate beginners of technical writing.
Permalink: <http://proxy-remote.galib.uga.edu/login?url=http://search.ebscohost.com/login.aspx?direct=true&db=nlebk&AN=28799&site=eds-live>

Answers questions like:

- Why technical writing exists (Chapter 1)
- How to identify the purpose and audience of technical writing (Chapter 5)



The University of Georgia®

ELEE 4230/6230: Sensors and Transducers

Course Project – Semester 20XX

A FUN TITLE FOR YOUR COURSE PROJECT

YOUR NAME

Insert a representative photo of your finished course project

Abstract

Write a short, structured abstract in this order: background, objective, methods, results, and conclusions. After reading the abstract, the reader should be clear on topic, goals, a brief description of the project, and what results were obtained and if these results were expected. (<https://www.enago.com/academy/abstract-versus-introduction-difference/>)

Problem Introduction

Do's

- Introduce the project. Yes, you can use 1st person writing to discuss why you are interested in it.
- Be clear about what problem you are trying to solve (not what solution you have).
- Include any photos need to help describe the big picture of what you are trying to accomplish.
 - The best problems often require multiple sensors.
 - For example, the problem could be **creating “the best cup of coffee”**. This is the problem. It does not have direct solution. You need to define and justify what parameters create “the best cup of coffee”.
- Be sure to use all the sensors vocabulary from the first few lectures correctly.
- Be sure to discuss what needs to be measured or solved.
- Include ideas of loading errors and other relevant sensors terminology.
 - I.e. let's say you are measuring current, any sensor will add resistance to the system and therefore, $V=IR$, so you will lose current in the measurement. This is a theoretical discussion.

Don'ts

- Do not state how you have solved the problem. For example, a problem might be measuring a distance. Do not state that “*you used an ultrasonic sensor with a microcontroller to measure distance.*” Here the problem is the focus, not the solution.
- Go past 1-2 pages.

Physical Environment and Project Constraints

Do's

- In the previous section you described the problem. Now you need to fully describe the environment that you are measuring.
- Include how the constraints of the project shape the selection.
 - Include all constraints...
 - Cost, time, effort, a good datasheet
 - Availability, longevity
 - Possible research or lab goals (i.e. grad student research)
- The environment you are measuring in determines what sensors are acceptable.

- At the end of this section the reader should be clear what environmental ranges a sensor would need to adequately operate in.
- This analysis should also influence what you need to measure to solve the problem you introduced.

Don'ts

- Skip the sensors safety checklist. You should complete the first 4 pages of the course document: General Safety Checklist for Sensors. This should be included in the Appendix.
- Be ignorant about the environment. Most sensors failures occur because some environmental factor that was unmodeled overwhelmed the measurement.
- Be clear about what is modeled and not modeled and how this will affect testing.
- Again, you have not chosen sensors yet. Building on the example, if you are measuring distance and you are choosing that to be the measurement that is redundant you shouldn't state how you are doubly measuring distance yet.
- Go past 1-2 pages.

Sensor Selection

Do's

- Explain how the desired problem to be solved, the physical environment, and the project constraints are going to be solved. What types of sensors are you going to use? Connect this back to the original problem you sought to go after.
- Start big then go small. For example, if you are measuring temperature, create a table listing many options then make pros/cons based on YOUR project.
- Make a final list of sensors used and rationale for why they are the best for this project. You can get to a final list of sensors and specs on the sensors and how that will meet the needs of the project.

Don'ts

- Forget to tie your selection into downstream requirements. Building on the distance measurement example, let's assume we have a requirement to measure 0.1". That can influence the sensor selection.
- Go past 1-2 pages.

RAW Sensor Dynamics and Signal Processing

Your documentation should follow the signal processing path used in nearly every lecture. Use subheadings in your writing to address the following topics:

- *Raw Physics*
 - Include a description of what the sensitive element is.
 - Include a graphical description of the sensors' response is.

- Include any graphics describing the physics of motion.
- Mathematical Model chosen (pros/cons)
 - The raw physics should include a graph of the raw physics. Here you need to justify what mathematical model you are using to describe the response.
 - Include what your mathematical model does NOT include (likely you have a single input, single output algebraic transfer function).
 - Include pros and cons. For example, you may be able to have a linear function if you accept error. Here you are justifying your mathematical model.
 - You will also need to discuss the range of the mathematical model.
- *Transducers* (if applicable)
 - If your system has transducers, you should discuss how the signal is transduced here.
- *Signal processing options*
 - Now that you have a mathematical model you need to discuss OPTIONS for how the signal is getting to a voltage or current that will be scaled appropriately for your microcontroller.
 - A picture is really worth 1000 words here. A graph can go a long way in helping you decide what signal processing you should choose.
 - You need to tie your signal processing choice back to requirements. For example, a huge constraint is that it needs to be active.
- *Power options*
 - Likewise list what power options you have for the project.
 - Give pros/cons of each option.
- *Final Signal Processing Chosen*
 - Yay! You're finally here! Include a circuit of your final signal processing and power options! Include graphs and figures showing how it is SHAPING the signal.
- *Filtering Chosen*
 - Include a discussion of what noise you should expect in the system. Measurements and real data are far more convincing fyi.
 - Include the circuit topology of the filter chosen and rationale for why that filter was chosen.
- *Microcontroller Code*
 - Since you've written the RAW sensor as a subfunction, explain how this subfunction goes "down the signal processing mountain" to bring the measurement back to the original value.
 - INCLUDE HOW THE FINAL VALUE IS CALIBRATED or what it is calibrated to.
 - What is the final resolution you can get out of the measurement?
- *Kalman Filter Used*

- Since this is the raw measurement, most people put the Kalman filter on this measurement. Explain the algorithm used to process this measurement.
- *Video results*
 - Finally, add a video here (or upload to ELC) showing the raw signal measurement.

Other Sensors Used

For all of the ‘pre-packed’ sensors duplicate this algorithm “N” times.

- Describe the raw physics of the sensor.
- Create a guess of the signal processing used in the sensor.
- Describe the output of the sensor. Include information on range, resolution, accuracy, etc.
- Include the library used for the sensor and what it is processing.

Digital Logic Implementation

Do’s

- Give a description of the list of the minimum set of requirements your microcontroller would need to realize this project.
- Give justification for the microcontroller you used in the project.
- Give a description of all algorithms used.
 - You’ve already described how the sensors get processed. Include the IOT or other algorithms used.

Don’ts

- Paste code in.
- Fail to give a description of what the code is doing.
- Go past 1-2 pages.

Final Design and Experimental Results

- 5 pages maximum
- If you are going for more than 75%
 - Include a photo and describe the 3D print.
 - Include a technical description of the battery setup used.
 - Include PCB designs as available.
- All submissions
 - Include a video in the submission of the final demonstration

- Include intermediate steps showing the signal processing realized for the op-amp setup.
- The author should present the results unbiased from personal input. Then create a subsection for 'Analysis of the Results'. Here you can add your interpretations about the results.

Conclusions

- 2 page maximum
- Improvements for a Revision 2.
 - If you were to do it all over again, what would you change?
 - If your instructor was to assign this all over again, what would change?
- Lessons Learned
 - You can break 3rd person writing here 😊

Human References

- Include references

AI References

- Give a list of AI tools and information gleamed from them.
- Remember you should give a prompt example when citing.

Appendix

- As long as needed